

Resistors



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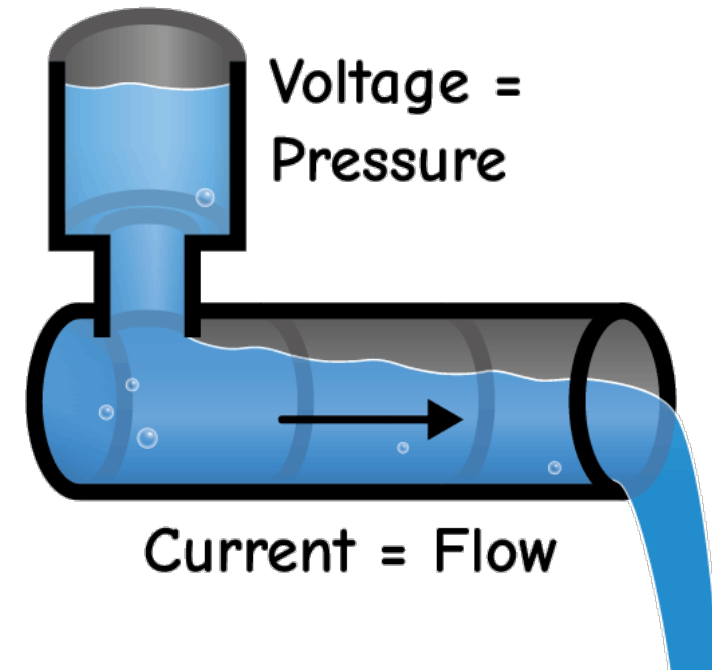
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Resistance

- Resistance is the opposition to current flow
- It has the symbol R with the unit Ohm (Ω)

Analogy

- Like a **narrow pipe** restricting water flow.
- Factors affecting resistance:
 - Material (copper vs rubber)
 - Length (longer wire = more resistance)
 - Thickness (thicker wire = less resistance)
 - Temperature (hotter wire = more resistance)

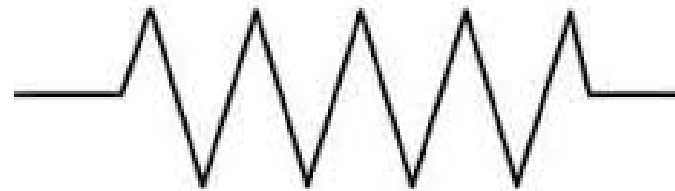


What is a resistor

- A resistor is an electrical component that opposes the flow of electric current.
- It creates a voltage drop when current passes through it.
- Resistance is measured in ohms (Ω).
- Purpose: to control current, divide voltages, and protect components.



As per IEC
standard



As per
American
Standard



Function of a Resistor

Current Limiting

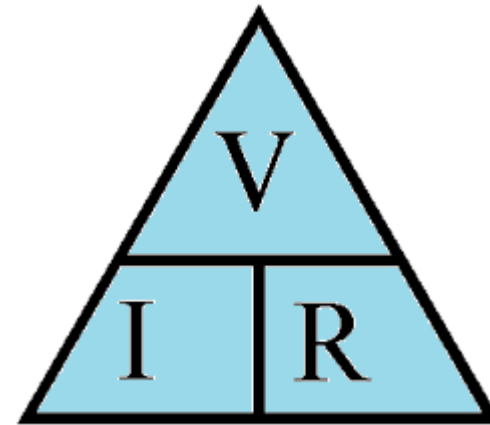
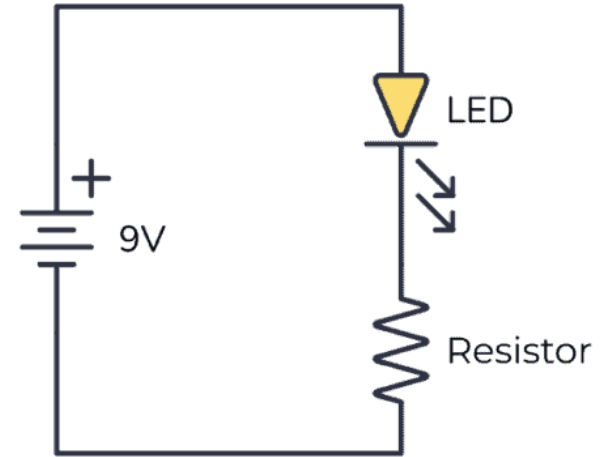
Voltage Division

Signal Conditioning

Heat Dissipation

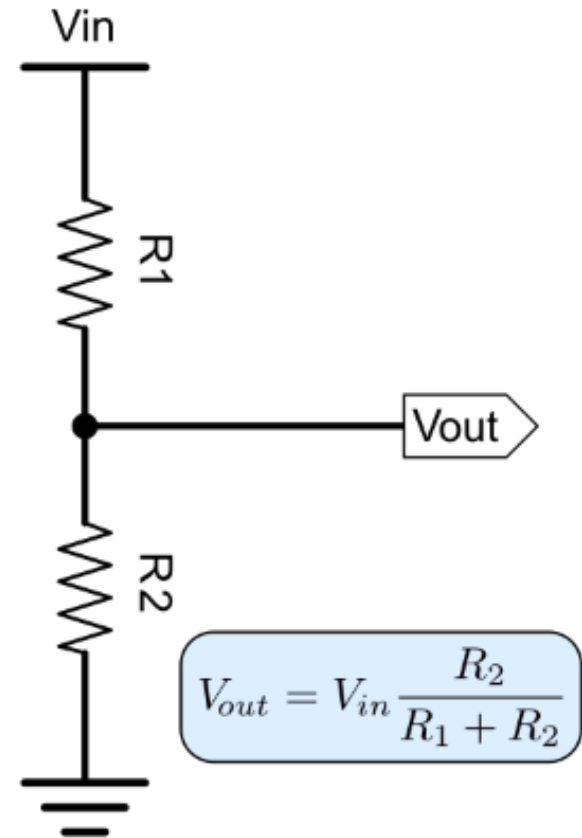
Current Limiting

- **Purpose:** Protect components from excessive current.
- **Example:** LEDs need current limiting to prevent burning out.
- **How it Works:** The resistor restricts current according to Ohm's Law
- **Typical Application:** Resistor in series with an LED.
- **Key Point:** Without a resistor, sensitive components can be damaged.



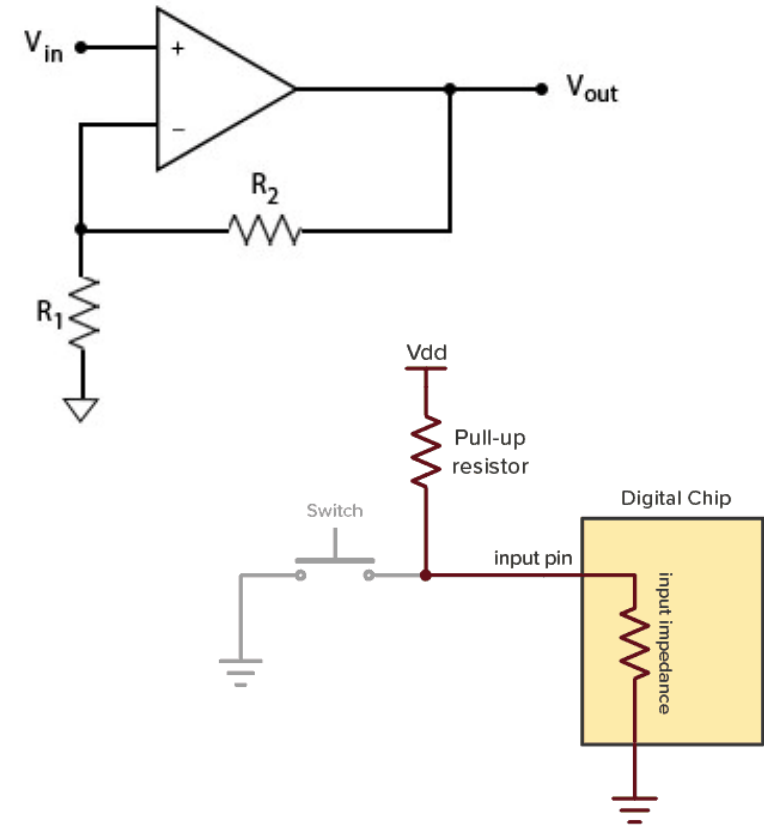
Voltage (potential) Divider

- **Purpose:** To split an input voltage into a smaller output voltage.
- **Circuit:** Two resistors in series across a supply.
- **Applications:**
 - Sensor circuits (e.g. LDRs, thermistors)
 - Reference voltages
 - Scaling signals for measurement



Signal Processing

- **Biasing:** Set operating points in amplifiers (e.g. transistor base bias).
- **Pull-up/Pull-down:** Ensure logic inputs default to a known state (0 or 1).
- **Filtering (with capacitors):** Form RC low-pass or high-pass filters. Control which frequencies are allowed through.
- **Impedance Control:** Match circuit stages to prevent signal loss or distortion.



Heat Dissipation

- **Why it matters:** Resistors convert unwanted electrical energy into heat.
- **Power Rating:**
 - Small resistors: ¼ W – 1 W
 - Wirewound resistors: up to 100 W+
- **Applications:**
 - Used as heaters (to burn off excess power)
 - Protection in high-current circuits
- **Key Point:** Always choose a resistor with a power rating higher than expected dissipation.

$$P = VI = I^2R = \frac{V^2}{R}$$

Types of Resistors

- We can divide resistors 3 main ways:

- **Fixed**

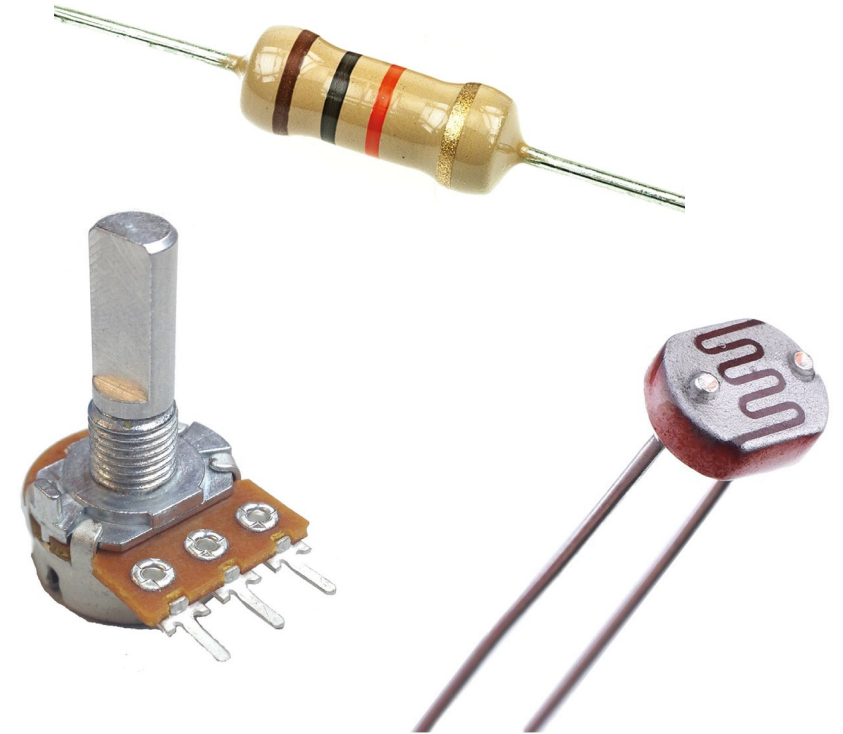
- Carbon Film
- Metal Film
- Wire Wound

- **Variable**

- Potentiometer
- Rheostats

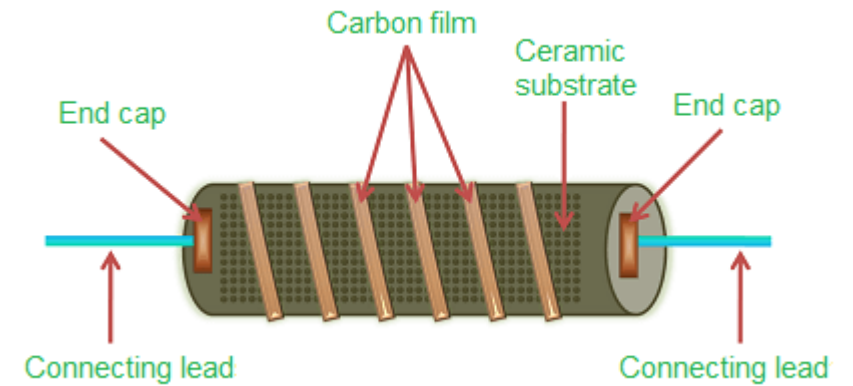
- **Special variable resistors**

- LDR
- Thermistor
- Varistor



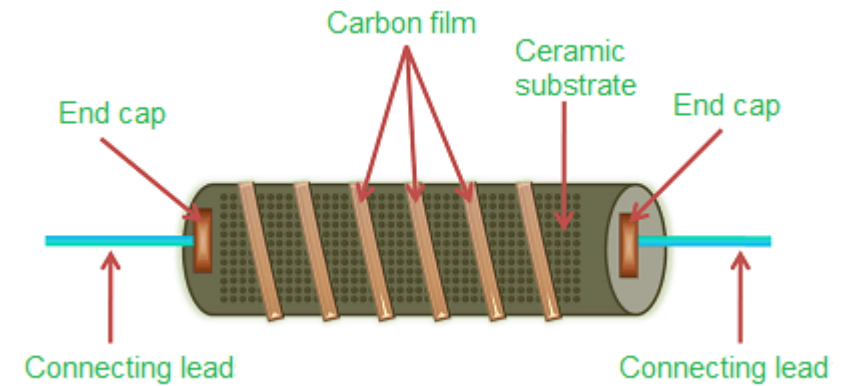
Carbon Film Resistors

- **Construction:** A thin film of carbon deposited on an insulating substrate.
- **Resistance Value:** Controlled by the thickness and length of the carbon film.
- **Power Rating:** Typically, low to medium ($\frac{1}{4}$ W to 2 W).
- **Tolerance:** Around $\pm 5\%$ (standard), but can be tighter.



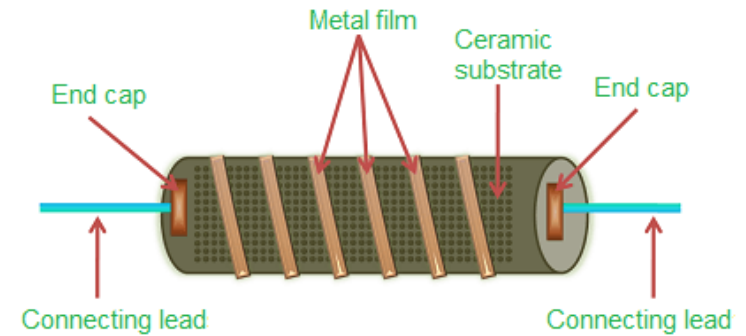
Carbon Film Resistors

- **Advantages:**
 - Cheap and widely available
 - Good stability for general use
- **Limitations:**
 - Higher noise than metal film resistors
 - Not suitable for high precision applications



Metal Film Resistors

- **Construction:** Thin layer of metal (often nickel-chromium) deposited on a ceramic rod.
- **Resistance Value:** Adjusted by cutting a helical groove in the film.
- **Power Rating:** Typically low ($\frac{1}{8}$ W to 1 W).
- **Tolerance:** Very precise, often $\pm 1\%$ or better.



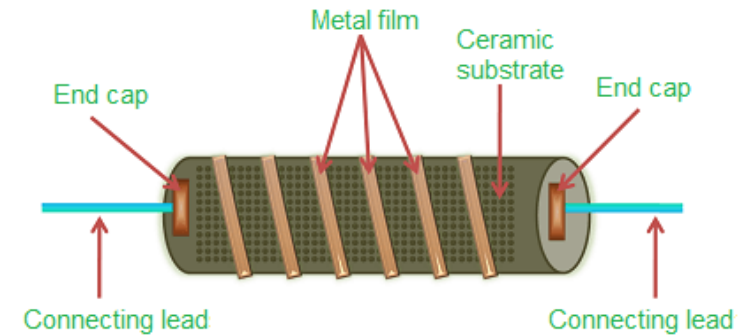
Metal Film Resistors

- **Advantages:**

- High accuracy and stability
- Low noise compared to carbon film
- Good temperature performance

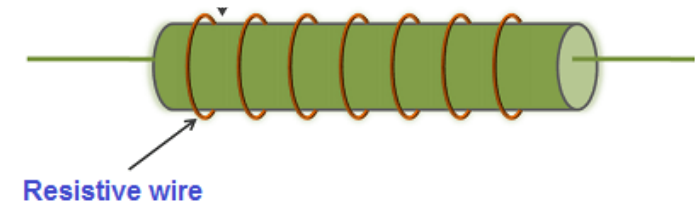
- **Limitations:**

- Slightly more expensive than carbon film
- Limited to low–medium power applications



Wire wound Resistors

- **Construction:** Resistive wire (usually nichrome) wound around a ceramic or fiberglass core.
- **Resistance Value:** Determined by the length and thickness of the wire.
- **Power Rating:** High – can handle several watts to hundreds of watts.
- **Tolerance:** Precise, typically $\pm 1\%$ or better.



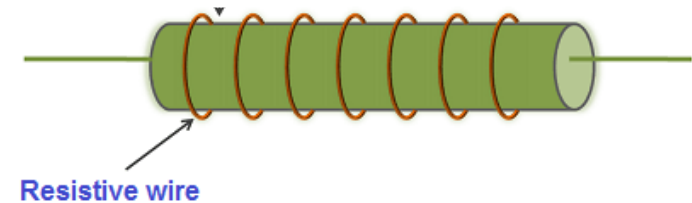
Wire wound Resistors

- **Advantages:**

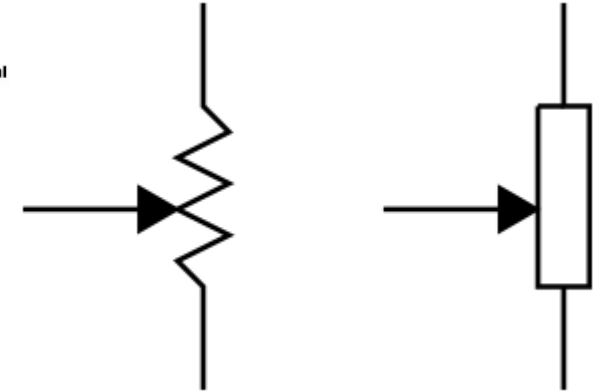
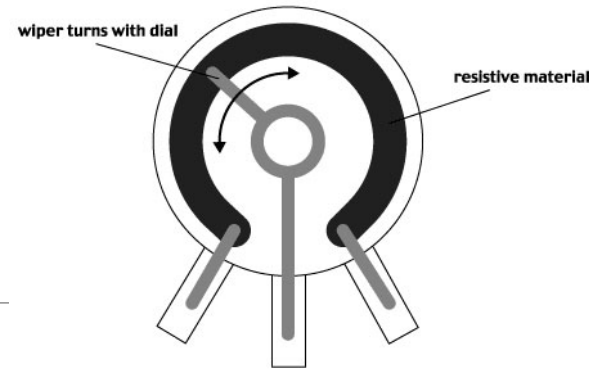
- Excellent stability and accuracy
- Can dissipate large amounts of power
- Low temperature coefficient

- **Limitations:**

- Larger physical size
- Inductive properties (not ideal for high-frequency circuits)
- More expensive than film resistors



Potentiometers



- **Potentiometer (3 terminals):**
 - Adjustable resistor with a wiper.
 - Used as a voltage divider (all 3 terminals).
 - Common in volume knobs, sensor calibration, etc.

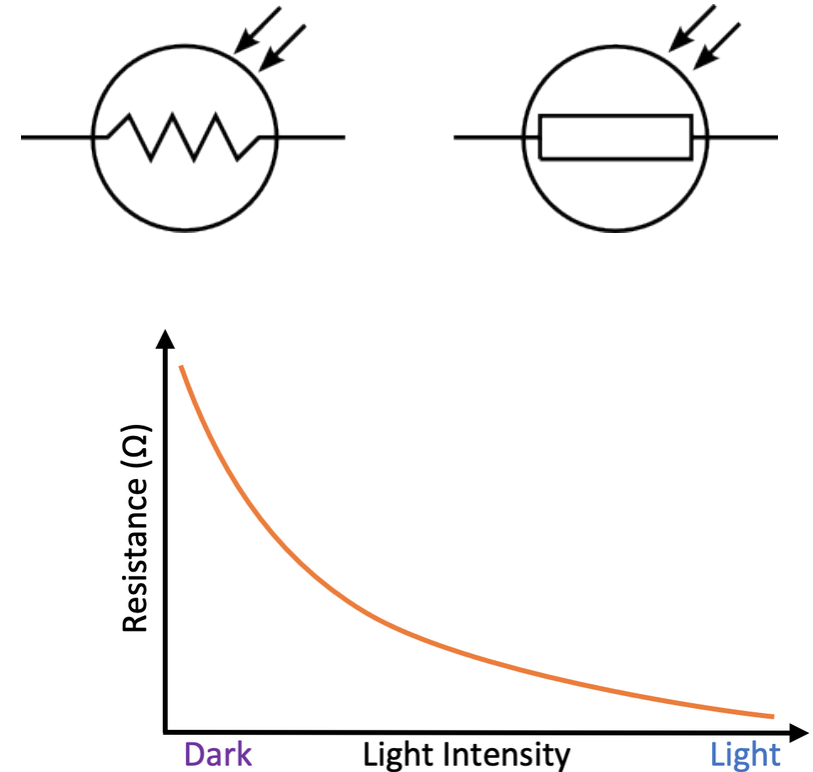
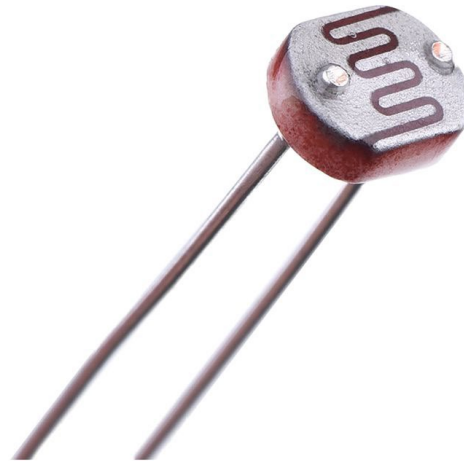
- **Rheostat (2 terminals):**
 - Potentiometer used as a variable resistor.
 - Controls current directly.
 - Common in lamp dimmers, motor speed control.

- **Types:**
 - Rotary (knob style)
 - Linear (slider style)



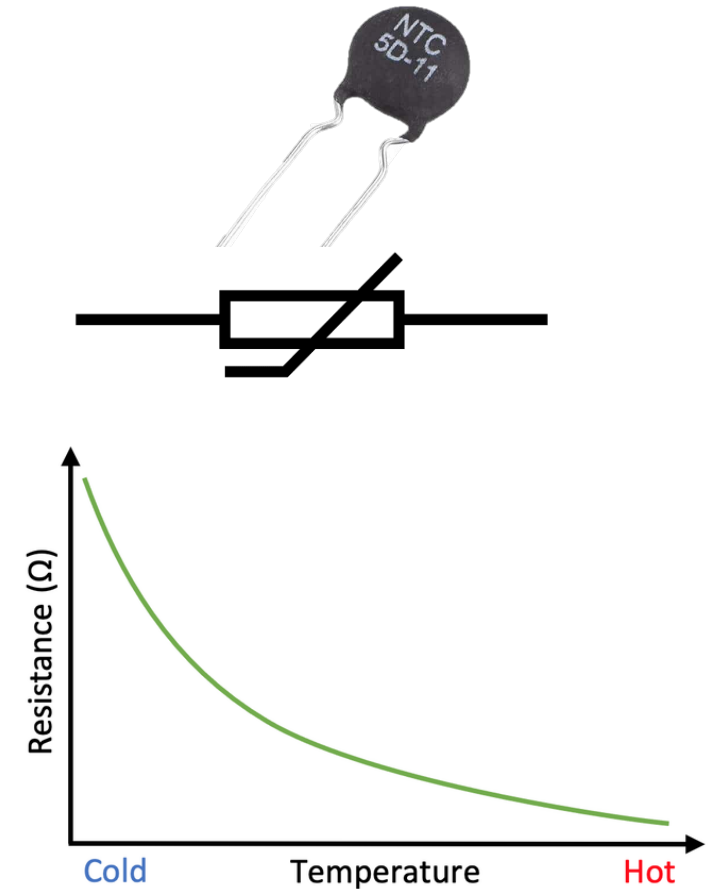
Light Dependent Resistors (LDRs)

- **Definition:** A special type of variable resistor whose resistance changes with light intensity.
- **Behaviour:**
 - Bright light → low resistance
 - Darkness → high resistance
- **Applications:**
 - Automatic street lights
 - Light meters (cameras)
 - Solar garden lights
- **Key Point:** Often used in a voltage divider to convert light levels into a voltage signal.



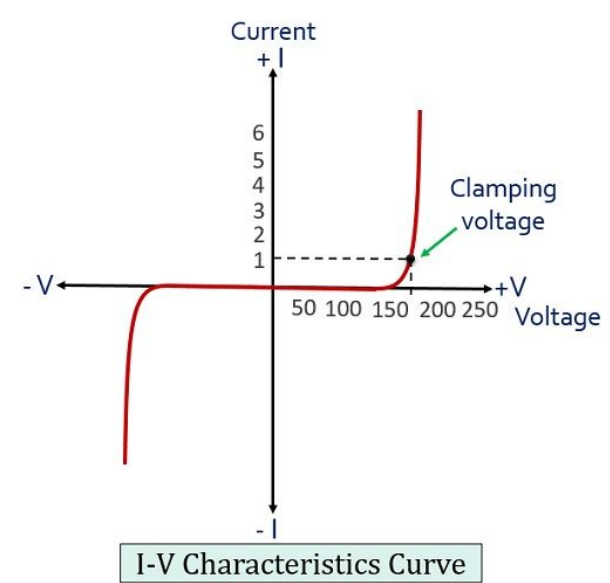
Thermistor

- **Definition:** A type of resistor whose resistance changes with temperature.
- **Types:**
 - NTC (Negative Temperature Coefficient): Resistance decreases as temperature increases. Common in temperature sensors.
 - PTC (Positive Temperature Coefficient): Resistance increases as temperature increases. Used in resettable fuses and overcurrent protection.
- **Applications:**
 - Digital thermometers
 - Temperature compensation in circuits
 - Overheat protection in power supplies
- **Key Point:** Often used in a voltage divider like an LDR but responds to heat instead of light.

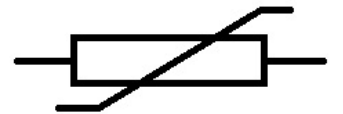


Varistor

- **Definition:** A resistor whose resistance changes with applied voltage.
- **Common Type:** MOV (Metal Oxide Varistor).
- **Behaviour:**
 - At normal voltages → very high resistance (almost open circuit).
 - At high voltages → resistance drops sharply, clamping the voltage.
- **Applications:**
 - Surge protection (e.g. in power strips, appliances).
 - Protecting circuits from voltage spikes (lightning, switching surges).
- **Key Point:** Acts like a safety valve – only conducts when voltage exceeds a set threshold.



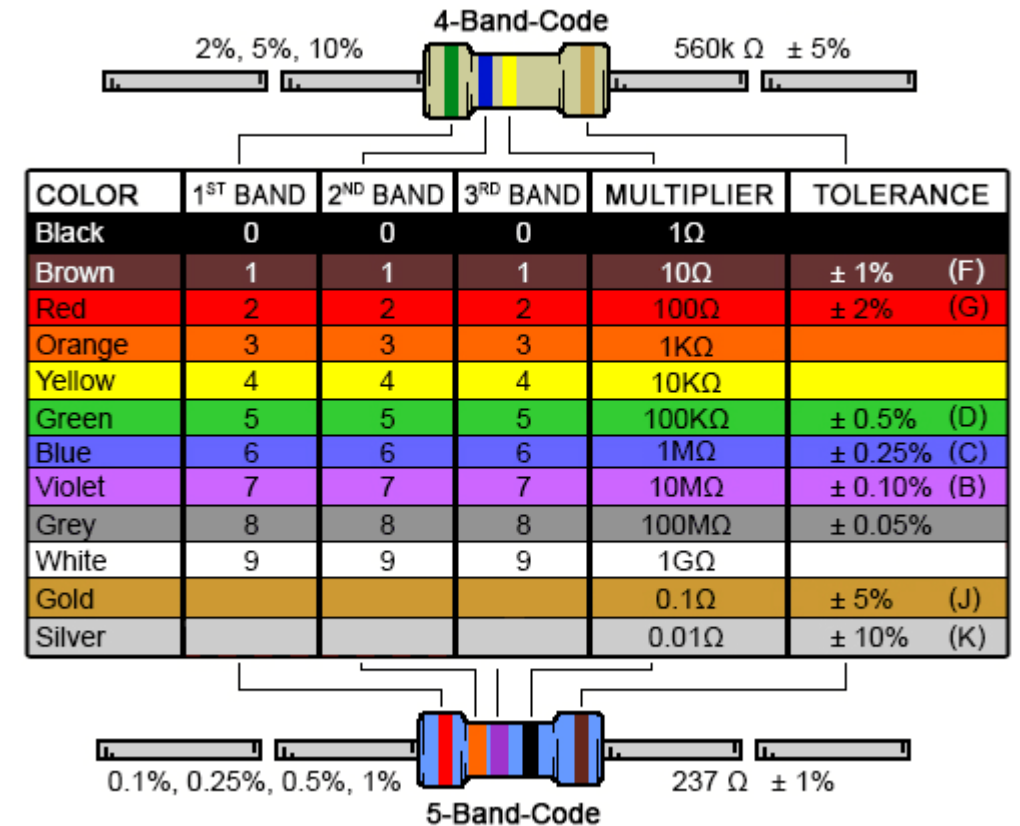
IEEE Standard Symbol for Varistor



IEC Standard Symbol for Varistor

Identifying Resistors

- **Resistor Colour Code:**
 - Each band represents a number or multiplier.
 - Final band = tolerance (gold = $\pm 5\%$, silver = $\pm 10\%$).
- **Other Identifiers:**
 - Marked values (printed on some precision resistors).
 - Case size indicates power rating ($\frac{1}{4}$ W, $\frac{1}{2}$ W, etc.).



Identifying Resistors - Example

2%, 5%, 10%

4-Band-Code

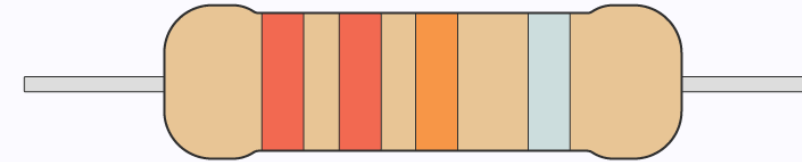
560k Ω $\pm$ 5%

COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1 Ω	
Brown	1	1	1	10 Ω	$\pm$ 1% (F)
Red	2	2	2	100 Ω	$\pm$ 2% (G)
Orange	3	3	3	1K Ω	
Yellow	4	4	4	10K Ω	
Green	5	5	5	100K Ω	$\pm$ 0.5% (D)
Blue	6	6	6	1M Ω	$\pm$ 0.25% (C)
Violet	7	7	7	10M Ω	$\pm$ 0.10% (B)
Grey	8	8	8	100M Ω	$\pm$ 0.05%
White	9	9	9	1G Ω	
Gold				0.1 Ω	$\pm$ 5% (J)
Silver				0.01 Ω	$\pm$ 10% (K)

0.1%, 0.25%, 0.5%, 1%

5-Band-Code

237 Ω $\pm$ 1%



Red	Red	Orange	Silver
2	2	10 ³	±10%

22000 Ω $\pm$ 10%